Ghazi Muhammad Abdullah

Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

Introduction

This report summarizes the work completed for **Experiment 06: Unsupervised Learning – Clustering and Principal Component Analysis (PCA)** as part of the M-Tech Production & Marketing AI/ML Internship Program 2026. The purpose of this experiment was to understand how unsupervised Machine Learning techniques identify hidden patterns in unlabeled datasets. During this experiment, I learned how to apply the K-Means clustering algorithm, determine the optimal number of clusters using the Elbow Method, and perform dimensionality reduction using Principal Component Analysis (PCA) with the Scikit-learn library.

Objectives

- Understand the fundamentals of unsupervised learning.
- Learn how the K-Means clustering algorithm groups similar data points.
- Determine the optimal number of clusters using the Elbow Method.
- Understand the concept and purpose of Principal Component Analysis (PCA).
- Learn how to reduce high-dimensional datasets into fewer dimensions.
- Visualize clustered and transformed data for better interpretation.

Task Description

This experiment focused on the concepts of unsupervised learning, where Machine Learning algorithms analyze data without predefined labels. I studied how the K-Means clustering algorithm groups similar data points into clusters and how the Elbow Method helps identify the most suitable number of clusters. I also learned how Principal Component Analysis (PCA) reduces the number of features in a dataset while preserving the most important information, making data easier to analyze and visualize.

Work Done

During this experiment, I studied the concepts of unsupervised learning and explored how clustering algorithms organize similar data into meaningful groups. I learned the working principle of the K-Means algorithm and applied it to sample datasets to observe how data points are assigned to different clusters. I also practiced using the Elbow Method to determine the optimal number of clusters by analyzing inertia values.

Furthermore, I studied Principal Component Analysis (PCA) and understood how it reduces the dimensionality of datasets while preserving most of the useful information. I applied PCA to transform a multi-feature dataset into two principal components, making it easier to visualize and analyze. I also observed how clustering and dimensionality reduction techniques are useful in customer segmentation, exploratory data analysis, and other real-world Machine Learning applications.

Challenges Faced

Initially, understanding how clustering algorithms work without labeled data was slightly challenging. Interpreting the Elbow Method graph to determine the optimal number of clusters also required careful observation. In addition, understanding how PCA reduces the dimensions of a dataset while maintaining important information required additional practice. However, after completing the practical exercises, these concepts became much easier to understand.

Key Learnings

- Developed a clear understanding of unsupervised Machine Learning.
- Learned how the K-Means algorithm groups similar data points into clusters.
- Understood the purpose and application of the Elbow Method.
- Gained practical knowledge of Principal Component Analysis (PCA).
- Learned the importance of dimensionality reduction for data visualization and efficient model performance.
- Improved my ability to analyze and interpret unlabeled datasets using Machine Learning techniques.

Conclusion

This experiment provided valuable practical experience with unsupervised Machine Learning techniques. By studying and implementing K-Means clustering and Principal Component Analysis, I strengthened my understanding of data grouping, dimensionality reduction, and data visualization. The knowledge gained during this experiment will help me apply these techniques effectively in future Machine Learning and Artificial Intelligence projects.

Intern Name: Ghazi Muhammad Abdullah

Supervisor Name: Mr. Mohazin Zahid

Date: 09 July, 2026

Domain: AI/ML Intern

Intern Signature: